

Canadian Junior Math Olympiad 2024 — P1/5

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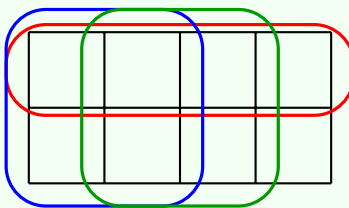
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Centuries ago, the pirate Captain Blackboard buried a vast amount of treasure in a single cell of a 2×4 grid-structured island. You and your crew have reached the island and have brought special treasure detectors to find the cell with the treasure. For each detector, you can set it up to scan a specific subgrid $[a, b] \times [c, d]$ with $1 \leq a \leq b \leq 2$ and $1 \leq c \leq d \leq 4$. Running the detector will tell you whether the treasure is in the region or not, though it cannot say where in the region the treasure was detected. You plan on setting up Q detectors, which may only be run simultaneously after all Q detectors are ready. What is the minimum Q required to guarantee your crew can determine the location of Blackboard's legendary treasure?

Solution

We claim that the minimum Q is 3. First we show a construction that works with $Q = 3$, then we show that it is impossible to have smaller Q .

Observe the following diagram,



setting up the detectors like so will guarantee that the crew is able to locate the treasure, since each square will alert a distinct set of detectors (this can easily be checked by hand).

Note that the only way to guarantee that the crew can determine the location of the legendary treasure is to have each square alert a distinct set of treasure detectors, because if we don't do this then there will be a set of squares that alert the same set of detectors and then the crew won't be able to guarantee knowledge of the location of the treasure. If the crew uses Q treasure detectors then there exists

$$\sum_{r=0}^Q \binom{Q}{r} = (1+1)^Q = 2^Q$$

different sets of treasure detectors that could be alerted. Since $2^1 < 2^2 < 2^3 = 8$, the minimum value of Q is indeed 3. ■

Remark: The intuition can come from computer science. It is well known that, given a sorted list L it is possible to search the list in $O(\log_2 n)$ time. Since there are $8 = 2^3$ cells in this case, it is natural to guess that a possible answer may be $\min Q = 3$.